

Mapping the Accessibility of Schools in Sarawak, Malaysia

Technical Notes

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Background

This document contains additional information on the methodology and caveats used in preparing the ‘Mapping the Accessibility of Schools in Sarawak, Malaysia’ blog post. It supplements the original post as a reference for those interested in scrutinising the assumptions, and/or reproducing this methodology.

Overview of Methodology

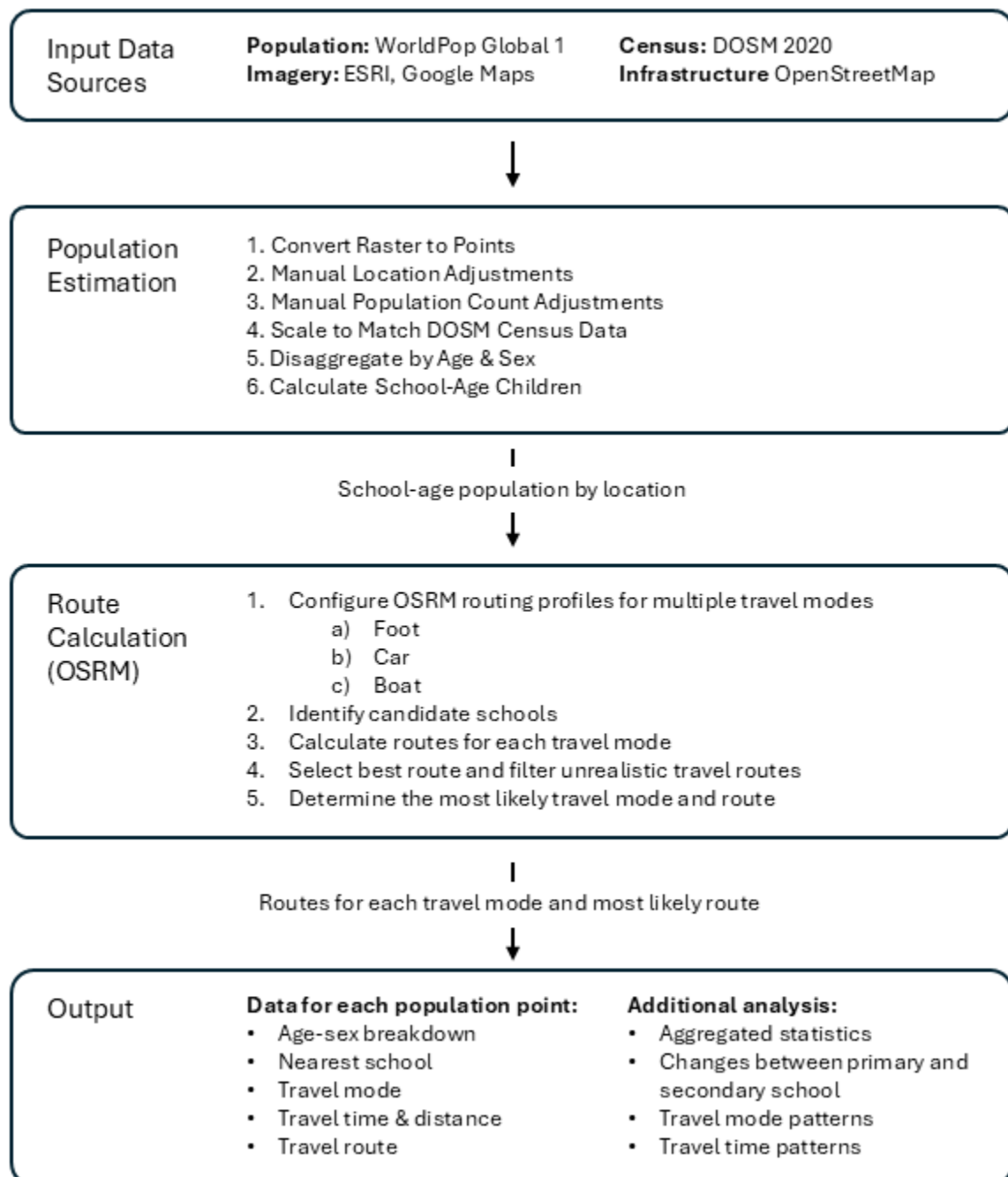


Figure 1: Methodology overview

Population Estimation (WorldPop)

WorldPop is a peer-reviewed, open access, project which aims to map out small population counts across the globe by using advance satellite imagery and machine learning techniques. It is developed by the School of Geography and Environmental Sciences at the University of Southampton. The data is and methodology is free to access via <https://www.worldpop.org/>.

A Brief Description of the WorldPop Methodology

WorldPop takes available census data and distributes it across a country based on where people are most likely to live. To do this, they:

1. Divide the globe into 100m x 100m grids
2. In each grid, data such as number of buildings, night-time light intensity, terrain, and land classification is compiled from various satellite imagery sources.
3. The machine learning models use these features to predict the likelihood of people living in that grid. Grids with many features suggests a high population density, whereas grids with no features suggests that nobody is living there.
4. These predictions are then calibrated with official sources where available (i.e. national census data and UN estimates).

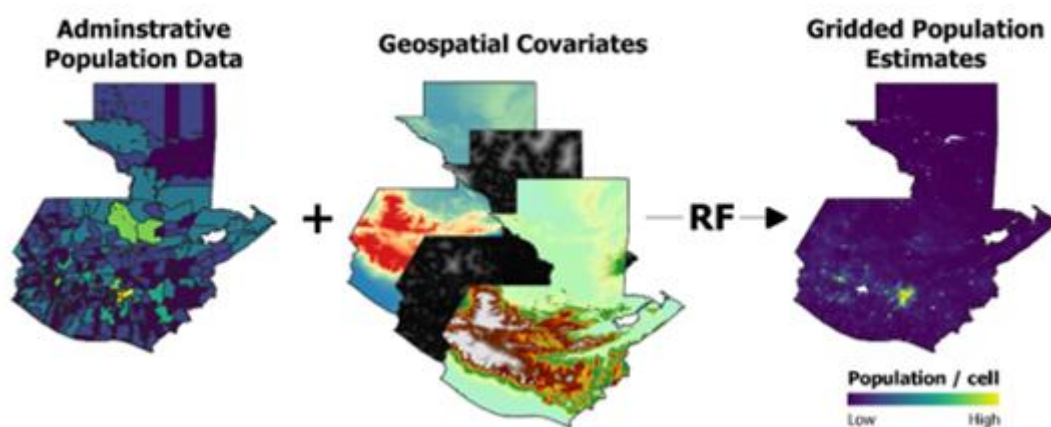


Figure 2: Schematic overview of the approach to generate gridded population estimates using the random forest (RF) model. For illustrative purposes, only a reduced set of considered covariates are shown here. (Source: WorldPop)

Full details on the methodology can be found on [WorldPop's website](https://www.worldpop.org/).

WorldPop Version

At the time of writing this study, two versions of WorldPop estimates were available. *WorldPop Global 1*, which was first published in 2018, and the latest *WorldPop Global 2*, which was released in September 2025. In general, *WorldPop Global 2* aims to provide more accurate estimates, because it utilises newer census and satellite data, as well as better-calibrated machine learning models.

This study uses *WorldPop Global 1* rather than *WorldPop Global 2*. This is because when this analysis first began, *Global 2* had not yet been released. By the time it was available, the study was already in relatively advanced stages, and switching datasets would have required redoing several months of work.

Although *WorldPop Global 2* could not be incorporated, the final dataset used in this study includes extensive calibrations applied to the *WorldPop Global 1* data (described later in this section). These calibrations would have been done regardless of the WorldPop version used and aims at improving accuracy, particularly in rural areas. As a result, using *WorldPop Global 1* instead of *Global 2* would likely have produced similar estimates.

Later in this section, the final *manually adjusted WorldPop* dataset is compared with both *WorldPop Global 1* and *WorldPop Global 2* wherever relevant. These comparisons are done to show that the *manually adjusted* dataset provides a more reliable and accurate estimate for the specific purpose of this study.

Manual Adjustments

No machine learning model is 100% accurate, so *WorldPop Global 1* is cross-checked against external sources. Two main manual adjustments are applied: **(1)** the locations of population points, and **(2)** the corresponding population count estimates.

Manually Adjusting Locations

WorldPop's raster data is first converted into point-level data, with each point representing a 1 km x 1 km grid cell on the map.

After that, the population point locations identified in *WorldPop Global 1* are plotted against recent [ESRI World Imagery](#) and [Google Maps Satellite Imagery](#) to verify their accuracy. Overall, *WorldPop Global 1* performs well in identifying settlements in urban and semi-urban areas, but it can miss remote longhouses or small settlements. In some cases, large plantations are misclassified as settlements, leading to population overestimation in areas where people are not likely to live.

To address these inaccuracies, population points are manually added, moved, or removed. A summary of these adjustments is given below.

Removing Points

WorldPop Global 1



WorldPop Manually Adjusted



Adding Points

WorldPop Global 1



WorldPop Manually Adjusted



Figure 3: Examples of population points being removed in a misidentified plantation, and points being added to remote longhouses that were missed. White points show the location of population points.

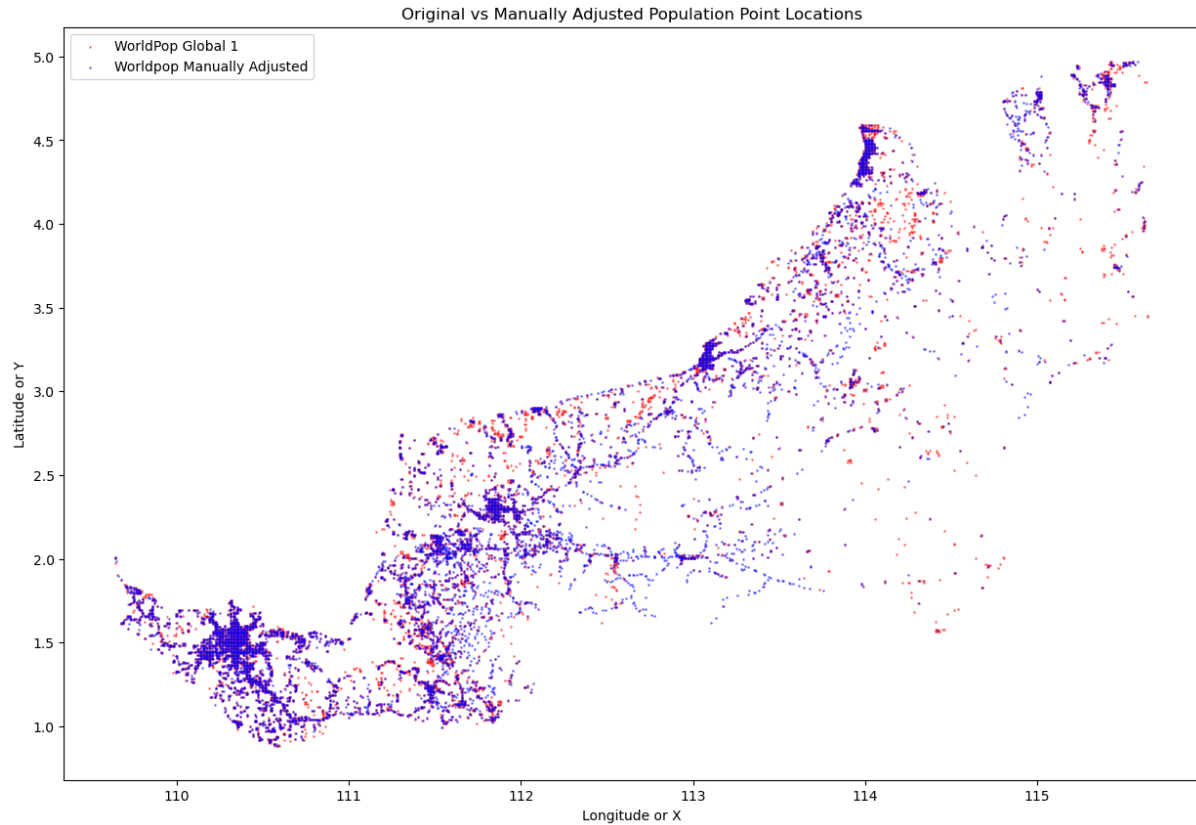


Figure 4: Comparing population point locations between WorldPop Global 1 and Manually Adjusted WorldPop.

Table 1: A summary of the original number of points, deleted points, moved points, and new points grouped by district

District	No. of Original Points	No. of New Points	No. of Moved Points	No. of Deleted Points
Overall Change	7965	1573	1851	1808
Asajaya	123	2	11	6
Bau	241	7	6	9
Belaga	207	25	119	63
Beluru	152	59	82	47
Betong	309	62	86	14
Bintulu	386	29	73	56
Bukit Mabong	83	69	45	30
Dalat	106	26	48	34
Daro	87	1	19	39
Julau	105	48	23	58
Kabong	110	18	10	57
Kanowit	130	93	19	76
Kapit	92	94	17	50
Kuching	766	20	19	49
Lawas	247	17	101	48
Limbang	128	55	38	35
Lubok Antu	257	58	81	126
Lundu	257	8	33	18
Marudi	128	28	75	35
Matu	138	7	65	37
Meradong	142	28	12	49
Miri	395	10	109	45
Mukah	211	49	93	67
Pakan	102	50	23	48
Pusa	65	2	20	0
Samarahan	185	3	11	12
Saratok	155	45	33	81
Sarikei	200	76	39	68
Sebauh	153	105	38	83
Selangau	150	101	45	74
Serian	341	11	22	2
Sibu	445	52	47	97
Simunjan	208	34	36	2
Song	68	75	28	31
Sri Aman	313	34	73	9
Subis	366	62	129	147
Tanjung Manis	57	3	15	14
Tatau	144	68	55	59
Tebedu	125	3	11	5
Telang Usan	88	36	42	28

Manually Adjusting Population Counts

After adjusting the locations, the population counts are also adjusted manually to match official census counts from the [Department of Statistics Malaysia \(DOSM\)](#) at a [district](#) and [DUN](#) level.

Manual changes to initial population estimates

First, the total population counts are manually adjusted alongside the initial process of manually adding, moving, or deleting population points from the *WorldPop Global 1* dataset. For existing points, values that are clearly under- or overestimated are manually changed. For newly added points, typically in rural longhouses or kampungs, a rule of thumb is applied: longhouses are assumed to roughly house 40-200 people, depending on their visible size. This provides a first-order estimate of population counts relative to other population points. A summary of these changes are made below.

Table 2: A summary of the original number of points, existing points with updated total population numbers and the average total population it was updated to, as well as the number of new points added and the average total population assigned to new points.

District	No. of Original Points	No. of Updated Points	Avg. Pop of Updated Points	No. of New Points	Avg. Pop of New Points
Overall Change	7965	407	156	1808	77
Asajaya	123	1	800	2	35
Bau	241	-	-	7	22
Belaga	207	10	153	25	97
Beluru	152	28	165	59	70
Betong	309	4	30	62	38
Bintulu	386	13	147	29	146
Bukit Mabong	83	11	101	69	74
Dalat	106	5	138	26	53
Daro	87	12	110	1	200
Julau	105	2	90	48	49
Kabong	110	3	32	18	45
Kanowit	130	13	176	93	71
Kapit	92	15	142	94	80
Kuching	766	-	-	20	26
Lawas	247	19	126	17	67
Limbang	128	12	138	55	71
Lubok Antu	257	17	78	58	32
Lundu	257	-	-	8	27
Marudi	128	25	161	28	78
Matu	138	9	141	7	34
Meradong	142	5	62	28	45
Miri	395	19	186	10	117

Mukah	211	22	186	49	93
Pakan	102	3	57	50	58
Pusa	65	-	-	-	-
Samarahan	185	-	-	-	-
Saratok	155	3	67	45	34
Sarikei	200	11	57	76	41
Sebauh	153	36	157	105	59
Selangau	150	4	120	101	74
Serian	341	1	400	11	60
Sibu	445	13	242	52	72
Simunjan	208	2	55	34	187
Song	68	1	50	75	64
Sri Aman	313	-	-	34	69
Subis	366	54	184	62	86
Tanjung Manis	57	2	250	3	343
Tatau	144	13	152	68	60
Tebedu	125	-	-	3	65
Telang Usan	88	19	192	36	90

Matching WorldPop data to DOSM Census data using the IPF method

Second, when aggregated to the district or DUN level, *WorldPop Global 1* (and *Global 2*) does not match DOSM population census figures, likely because WorldPop is calibrated to national rather than sub-national census totals. To address this, the initial population estimates are scaled up or down to align with district and DUN population data when aggregated. This is known as the [Iterative Proportional Fitting \(IPF\)](#) method.

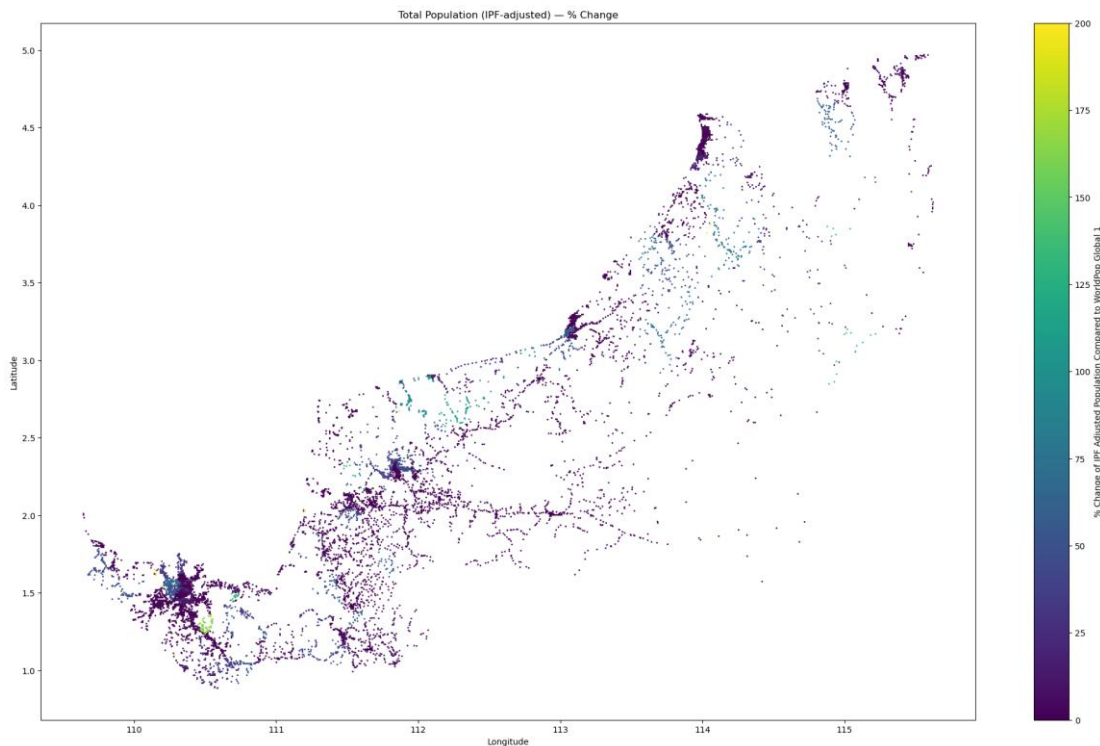


Figure 5: % Change in total population counts after applying the IPF method, when compared to WorldPop Global 1

Creating age breakdowns from the scaled total population counts

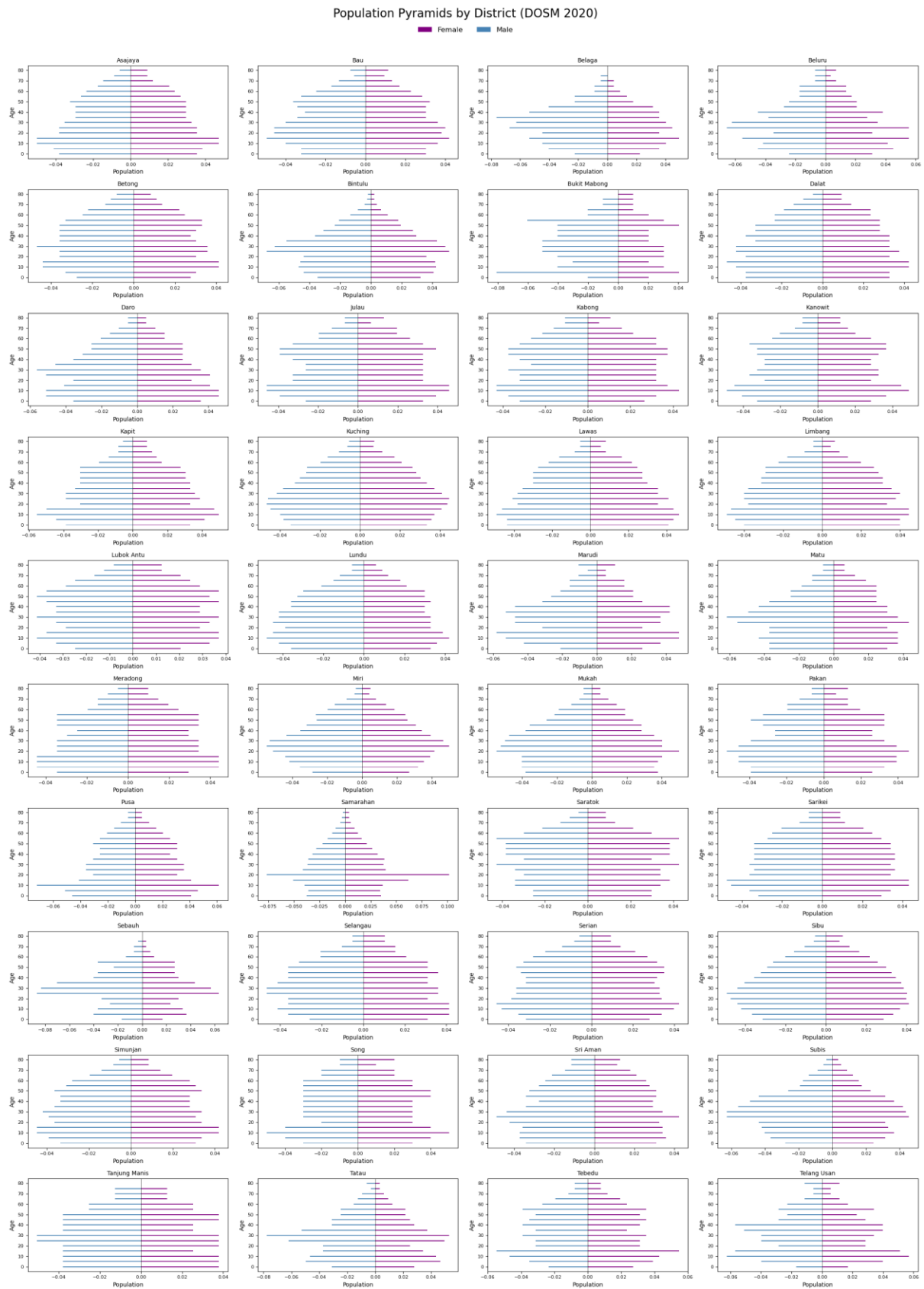


Figure 6: The age-sex pyramids used for disaggregation, based on DOSM's 2020 census data.

After total population counts are scaled, populations are disaggregated into age and sex groups using district-level demographic patterns. For each district, the proportion of the population by age and sex is first calculated using DOSM data. These proportions are then applied uniformly to every population point within the district by multiplying the point-level total population by the corresponding age- and sex-specific proportions.

Age-sex category breakdowns are calculated for every 5 years (e.g. 0-4 years old, 5-9 years old, 10-15 years old) for males and females.

Calculating the number of school-age children

Students are assumed to attend primary school between ages 7-12 years old, and secondary school between ages 13-17 years old. Because age-sex data are only available in 5-year bins, school-age populations are estimated proportionally from these bins. For example, the 7-12 age group is calculated as 60% of the 5-9 age bin plus 60% of the 10-15 age bin. The same applies for secondary schools.

Comparing WorldPop Global 1, Global 2, and Manually Adjusted

Comparing population point locations

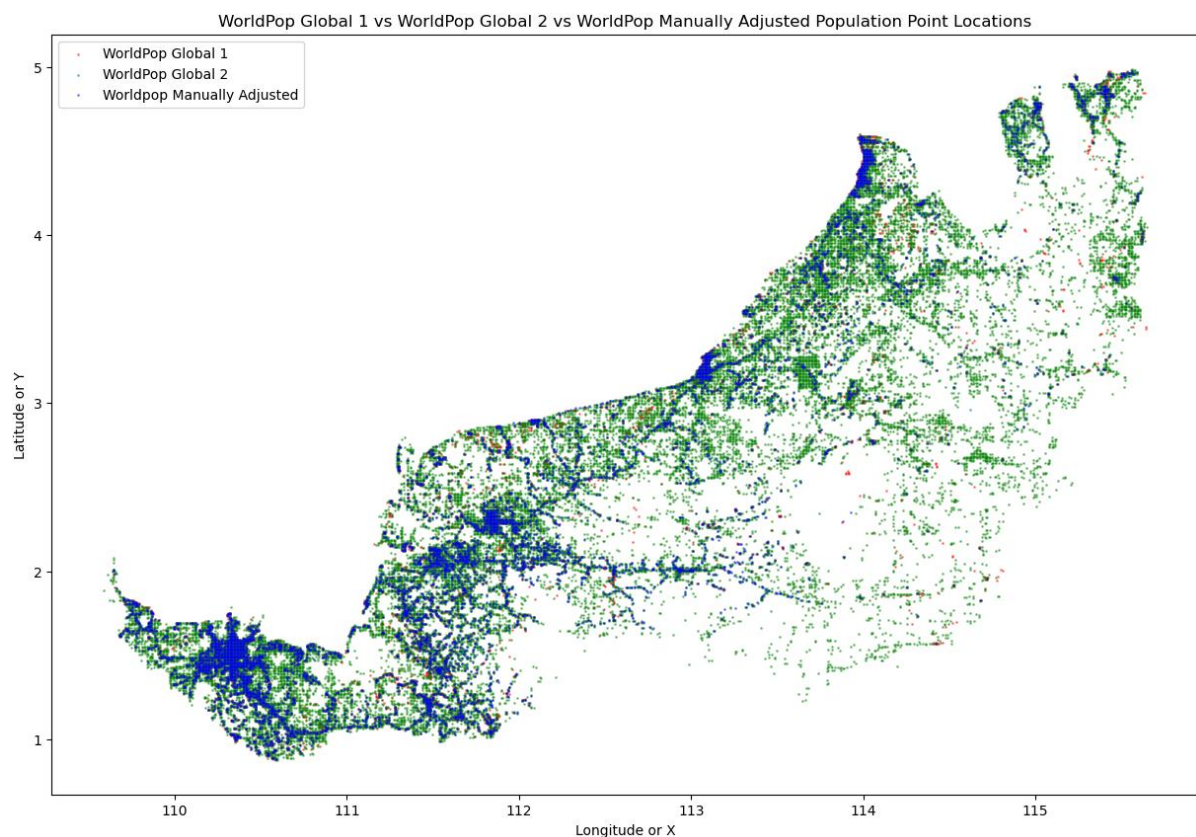


Figure 7: Population point locations of WorldPop Global 1 (red) vs WorldPop Global 2 (green) vs WorldPop Manually Adjusted (blue)

Comparing aggregated district and DUN data counts (thousands)

Table 3: A comparison of aggregated total population counts by districts.

District	WorldPop Global 1	WorldPop Global 2	WorldPop Adjusted	DOSM
Overall	2,677.5	2,521.6	2,466.2	2,454.4
Asajaya	32.0	34.9	33.9	33.9
Bau	61.9	53.8	52.6	52.6
Belaga	45.0	23.3	22.3	22.3
Beluru	12.8	27.7	28.9	28.9
Betong	41.3	37.0	36.3	36.3
Bintulu	192.4	189.1	178.5	178.5
Bukit Mabong	14.6	10.7	9.9	9.9
Dalat	12.1	19.0	21.3	21.3
Daro	28.5	20.3	19.6	19.6
Julau	13.9	15.6	15.3	15.3
Kabong	19.1	18.9	18.7	18.7
Kanowit	27.4	25.4	24.6	24.6
Kapit	41.1	37.2	36.1	36.1
Kuching	669.5	659.5	621.0	609.2
Lawas	39.4	38.3	36.7	36.7
Limbang	44.4	46.6	45.0	45
Lubok Antu	33.1	24.6	24.3	24.3
Lundu	35.8	33.8	33.4	33.4
Marudi	19.2	19.3	19.0	19
Matu	30.1	16.6	16.2	16.2
Meradong	25.2	20.9	20.3	20.3
Miri	296.4	253.8	248.9	248.9
Mukah	29.7	35.3	42.2	42.2
Pakan	13.9	15.8	15.3	15.3
Pusa	20.3	20.5	19.5	19.5
Samarahan	150.6	136.0	128.3	128.3
Saratok	26.0	23.9	23.5	23.5
Sarikei	49.5	45.0	44.1	44.1
Sebauh	31.4	33.1	29.9	29.9
Selangau	25.8	21.0	19.4	19.4
Serian	72.0	58.3	85.4	85.4
Sibu	248.1	256.3	248.0	248
Simunjan	35.4	36.2	35.7	35.7
Song	19.5	10.0	10.0	10
Sri Aman	64.0	61.9	61.5	61.5
Subis	66.1	57.0	57.3	57.3
Tanjung Manis	21.0	7.5	7.9	7.9
Tatau	35.1	32.9	32.2	32.2
Tebedu	23.5	25.4	25.6	25.6
Telang Usan	10.9	19.3	17.6	17.6

Table 4: A comparison of aggregated total population counts by DUN.

DUN	WorldPop Global 1	WorldPop Global 2	WorldPop Adjusted	DOSM
Overall	2,677.5	2,521.6	2,466.2	2,454.4
N.01 Opar	17.0	18.2	21.8	21.8
N.02 Tasik Biru	38.4	32.1	29.7	29.7
N.03 Tanjong Datu	23.7	17.7	16.0	16.0
N.04 Pantai Damai	41.5	51.2	56.0	56.0
N.05 Demak Laut	70.0	75.1	45.7	45.7
N.06 Tupong	54.6	51.1	91.3	91.3
N.07 Samariang	70.6	54.6	45.1	45.1
N.08 Satok	11.8	13.0	23.3	14.5
N.09 Padungan	17.5	23.2	17.2	14.2
N.10 Pending	58.4	52.0	37.1	37.1
N.11 Batu Lintang	53.5	62.5	65.0	65.0
N.12 Kota Sentosa	101.6	100.7	66.4	66.4
N.13 Batu Kitang	61.3	54.0	59.0	59.0
N.14 Batu Kawah	57.8	64.0	86.8	86.8
N.15 Asajaya	22.8	21.5	20.0	20.0
N.16 Muara Tuang	78.3	71.9	65.0	65.0
N.17 Stakan	73.8	69.3	73.5	73.5
N.18 Serembu	18.4	19.1	16.7	16.7
N.19 Mambong	49.1	38.4	38.2	38.2
N.20 Tarat	50.8	32.0	25.8	25.8
N.21 Tebedu	20.0	19.3	22.0	22.0
N.22 Kedup	15.0	16.8	18.0	18.0
N.23 Bukit Semuja	24.2	20.2	21.8	21.8
N.24 Sadong Jaya	9.9	11.6	11.0	11.0
N.25 Simunjan	7.5	12.2	12.3	12.3
N.26 Gedong	8.1	10.4	10.4	10.4
N.27 Sebuyau	10.6	9.7	11.8	11.8
N.28 Lingga	17.0	14.9	10.7	10.7
N.29 Beting Maro	16.5	16.0	15.8	15.8
N.30 Balai Ringin	20.7	17.9	15.2	15.2
N.31 Bukit Begunan	14.4	15.2	18.7	18.7
N.32 Simanggang	25.6	29.9	29.0	29.0
N.33 Engkilili	18.3	12.8	12.6	12.6
N.34 Batang Ai	16.3	13.5	14.2	14.2
N.35 Saribas	10.2	7.7	11.5	11.5
N.36 Layar	20.5	17.5	12.7	12.7
N.37 Bukit Saban	13.6	9.5	10.5	10.5
N.38 Kalaka	11.8	8.9	12.8	12.8
N.39 Krian	19.3	19.8	15.5	15.5
N.40 Kabong	13.4	12.9	13.1	13.1

N.41 Kuala Rajang	33.7	16.4	18.7	18.7
N.42 Semop	17.7	10.4	14.0	14.0
N.43 Daro	23.3	17.5	14.4	14.4
N.44 Jemoreng	23.4	12.8	13.0	13.0
N.45 Repok	23.2	25.3	24.2	24.2
N.46 Meradong	29.6	22.9	18.9	18.9
N.47 Pakan	14.8	18.1	17.2	17.2
N.48 Meluan	14.9	17.4	18.0	18.0
N.49 Ngemah	14.0	11.0	10.2	10.2
N.50 Machan	13.9	14.7	14.5	14.5
N.51 Bukit Assek	43.9	45.1	37.7	37.7
N.52 Dudong	75.5	59.3	67.3	67.3
N.53 Bawang Assan	20.8	29.6	26.6	26.6
N.54 Pelawan	50.3	44.9	42.0	42.0
N.55 Nangka	53.5	68.4	73.0	73.0
N.56 Dalat	10.8	14.5	16.7	16.7
N.57 Tellian	15.1	17.7	27.0	27.0
N.58 Balingian	14.9	19.3	17.1	17.1
N.59 Tamin	22.8	21.5	16.8	16.8
N.60 Kakus	40.3	34.9	34.5	34.5
N.61 Pelagus	14.1	10.7	11.9	11.9
N.62 Katibas	19.2	9.6	9.5	9.5
N.63 Bukit Goram	24.0	23.7	22.6	22.6
N.64 Baleh	8.0	8.7	9.4	9.4
N.65 Belaga	34.6	13.5	13.1	13.1
N.66 Murum	24.9	27.1	21.0	20.9
N.67 Jepak	31.9	33.0	35.3	35.3
N.68 Tanjong Batu	48.0	57.0	74.0	73.9
N.69 Kemena	57.9	40.3	47.2	47.1
N.70 Samalaju	122.4	106.8	78.8	78.7
N.71 Bekenu	21.3	26.2	22.1	22.1
N.72 Lambir	41.0	49.4	48.9	48.9
N.73 Piasau	40.0	26.3	34.6	34.6
N.74 Pujut	42.6	39.7	39.2	39.2
N.75 Senadin	175.5	141.0	132.5	132.3
N.76 Marudi	23.4	34.4	32.9	32.9
N.77 Telang Usan	12.0	20.7	16.0	15.9
N.78 Mulu	15.2	17.7	14.7	14.7
N.79 Bukit Kota	35.2	29.9	31.7	31.7
N.80 Batu Danau	9.2	14.1	13.3	13.3
N.81 Ba` Kelalan	11.4	16.0	9.7	9.7
N.82 Bukit Sari	28.3	21.9	27.0	26.9

Limitations

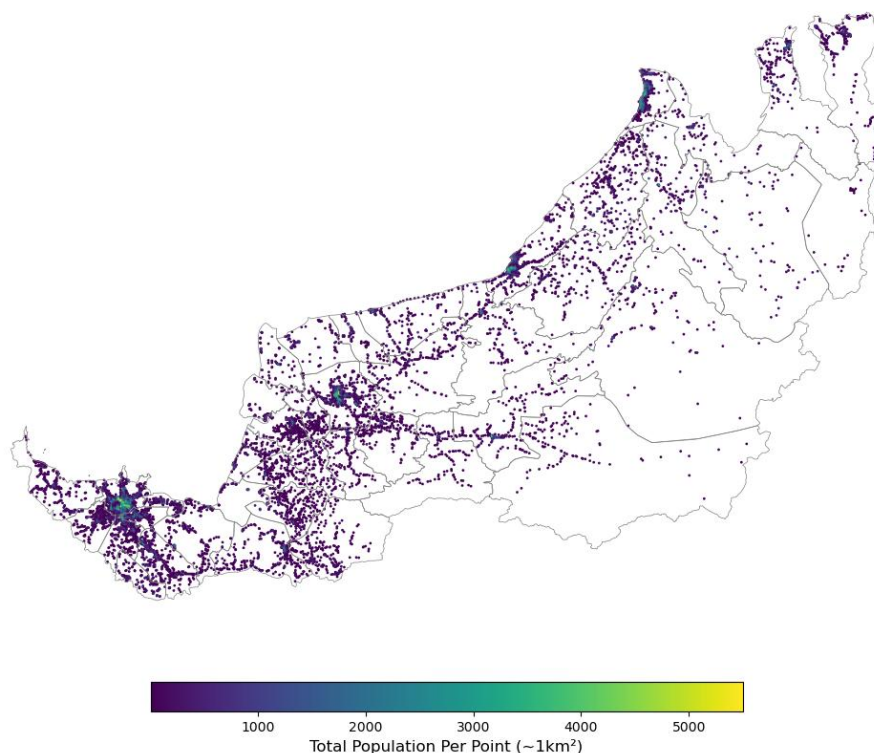
All forms of small-area population modelling are inherently uncertain and should be interpreted as broad patterns rather than exact counts. While every effort is taken to ensure the results are as representative as possible, errors may still arise from model assumptions, methodological limitations, or limits in researcher judgement. In addition, neither WorldPop or this analysis provides uncertainty estimates at the point level; wherever possible, results should therefore be interpreted alongside internal data or on the ground verification.

Because populations are disaggregated into age and sex groups, small population points may contain decimal values. For example, a point with a total population of 20 may be split into age-sex groups of only 0.1 or 0.2 people, which is not a meaningful result. This occurs because age-sex proportions are meaningful at an area level, but not at the level of individual households or points, where demographic composition varies substantially. For this reason, population estimates are always aggregated to the highest spatial and categorisation required for any analysis, before rounding is applied.

Final Population Counts

Having applied all these data processing methods, we finalise the small population count estimates as shown below.

Estimated Total Population in Sarawak in 2020



Route Calculation (OSRM)

A Brief Description on the OSRM Methodology

The [Open Source Routing Machine \(OSRM\)](#) is a fast, open-source routing engine, similar in function to Google Maps, Waze or Apple Maps. It uses [Open Street Map \(OSM\)](#) infrastructure data to calculate the fastest route between two points based on a given routing profile. For example, a car routing profile will only consider roads accessible by cars and exclude other paths. OSRM is widely used for large-scale distance, travel-time, and route calculations, and supports custom routing profiles (e.g. car, walking, cycling, boats), making it well-suited for this analysis.

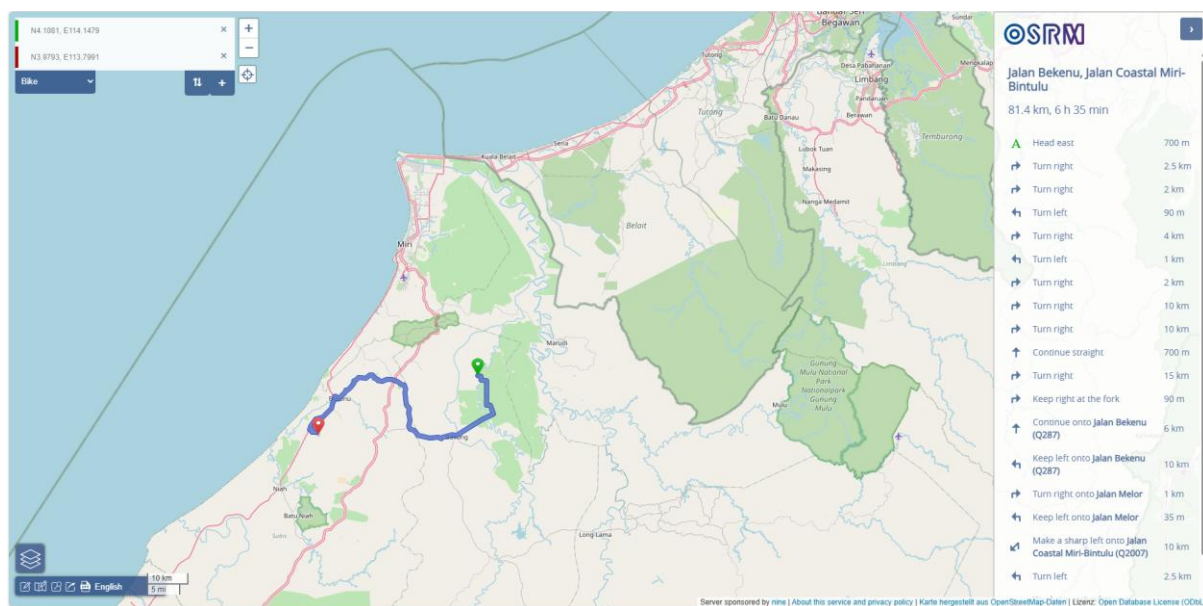


Figure 8: A sample online server is set up which can be used [here](#).

How to Set Up OSRM

For smaller tasks, OSRM can be accessed online via an API. However, this analysis requires custom routing profiles and many routing queries; hence, a locally hosted OSRM server is needed. Some useful sources on installing and using a locally hosted OSRM server is shown below:

- [OSRM documentation](#)
- [OSRM backend repository](#)
- [GitHub OSRM course](#)
- [YouTube OSRM course](#)

OSRM Routing Profiles

Once a local OSRM server is set up, we need to configure custom routing profiles. By default, OSRM provides routing profiles for cars, bicycles, and foot. For this analysis, we use the default foot profile, a customised version of the existing car profile, and a new boat profile. All routing profiles used are provided on [GitHub](#).

OSRM Foot

We use the default foot profile out-of-the box when installing OSRM. A brief, AI-generated summary of the foot profile:

The foot profile models pedestrian travel across all walkable OpenStreetMap features, assuming a baseline walking speed of **5 km/h**. It allows routing on most highway types, including primary to tertiary roads, residential and service roads, tracks, paths, footways, steps, pedestrian areas, platforms, piers, and living streets. Railway platforms, parking areas, and leisure tracks are also accessible for walking. Access is restricted where explicit prohibitions exist (e.g. *access=no*, *foot=no*, *private*, *forestry*, *agricultural*), and physical barriers such as walls or fences block traversal unless explicitly passable.

Surface conditions affect walking speed: fine gravel and gravel reduce speed to **75% of baseline**, while sand and mud reduce speed to **50%**, reflecting slower movement on poor terrain. Ferry routes are allowed at a fixed speed of **5 km/h**, enabling pedestrian river crossings where mapped. Traffic signals and U-turns introduce small time penalties, but motor-vehicle one-way restrictions are ignored unless explicitly tagged for foot traffic. Overall, the profile is designed to represent realistic pedestrian accessibility in rural and semi-rural contexts rather than urban footpath-only movement

Speed settings:

Table 5: Speed settings based on the type of road found in OSM for the foot profile.

OSRM tags	Speed Setting (km/h)
motorway	5
motorway_link	5
trunk	5
trunk_link	5
primary	5
primary_link	5
secondary	5
secondary_link	5
teritiary	5
teritiary_link	5
unclassified	5
residential	5
living_street	5
service	5
track	5

Other restrictions and conditions can be found in the “foot.lua” file in the “OSRM Profiles” folder.

OSRM Car (Customised)

For the car profile, we make minor adjustments to the OSRM profile to allow the use of roads classified as ‘track’ (which tend represent off-road or unpaved roads) when calculating the pathways. A brief AI-generated summary of the car profile:

The car profile models motorised travel using OpenStreetMap road infrastructure, routing vehicles across standard paved roads as well as unpaved tracks where permitted. It respects vehicle access rules, barriers, one-way restrictions, and basic vehicle constraints, ensuring that routes remain physically plausible.

Travel speeds are adjusted according to road type, surface material, track quality, and smoothness, with significant slowdowns applied to gravel, dirt, sand, mud, and poor-quality tracks. The profile uses routing penalties to favour higher-quality roads where available, while still allowing lower-quality routes when no alternatives exist. Overall, this customised car profile finds the best path available, considering restrictions and the quality of different paths.

Speed settings:

Table 6: Speed settings based on the type of road found in OSM for the car profile.

OSRM tags	Speed Setting (km/h)
motorway	10
motorway_link	25
trunk	85
trunk_link	40
primary	65
primary_link	30
secondary	55
secondary_link	25
tertiary	40
tertiary_link	20
unclassified	25
residential	25
living_street	10
service	15
track	30

Other penalties, restrictions, and speed limits by road surface type settings can be found in the “car_offroad.lua” file in the “OSRM Profiles” folder.

OSRM Boat

For the boat profile, a simple custom profile is written from scratch. A brief AI-generated summary of the boat profile:

The boat profile models motorised river transport using OpenStreetMap waterway data, routing only along features explicitly tagged as *river* or *canal*. All other road or land-based features are ignored. Rivers are assigned a higher travel speed than canals, reflecting typical differences in navigability and flow conditions. Routes are treated as bidirectional, allowing travel both upstream and downstream.

This profile does not attempt to model detailed hydrological effects, vessel types, or seasonal variability. Instead, it provides a simplified representation of inland waterway access, showing basic river reachability in regions where boats are a primary mode of transport.

Speed settings:

Table 7: Speed settings for the type of waterway found in OSM for the boat profile.

OSRM tags	Speed Setting (km/h)
canal	10
river	25

Routing Methods

For each travel mode we calculate the paths as follows:

1. For each population point, the three nearest schools are identified using straight-line (Euclidean) distance. This is implemented using the **cKDTree** method from the **scipy.spatial** Python library and serves as a computational pre-filter, reducing the routing problem from ~1,200 schools to a small set of plausible candidates.
2. Routing is then performed between each population point and its three candidate schools using OSRM. For each population–school pair, OSRM returns the fastest feasible route, including travel time, travel distance, and route geometry, subject to the assumptions of the selected routing profile.
3. The school with the shortest travel time is selected as the most likely school attended by each population point for each transport mode. If no valid route exists for any candidate school, the population point is classified as having no viable access.
4. Unrealistic routes are then identified and filtered¹:

¹ These inaccuracies were identified through extensive validation. While the selected values may appear somewhat arbitrary (and can be adjusted for sensitivity analysis) they were chosen after multiple rounds of testing to produce a representative set of results that reflects practical routes.

- a. If the ratio of OSRM route distance to Euclidean distance exceeds 10, this suggests that the initial candidate set may be too narrow. In such cases, routing is rerun using the 10 nearest schools by Euclidean distance, and the fastest valid route is selected.
 - b. Routes with excessive snapping distances (defined as more than 1 km for foot, 1.5 km for car, or 3 km for boat) are discarded, as they imply that the route is not physically reachable. The next-best route within these thresholds is selected instead.
5. The final routes are cleaned, validated, and saved.

Combined Routes

For each population point, routes are calculated for all three travel modes (boat, car and foot). For each mode, the model either returns the fastest route to the nearest school or no viable route.

While all routes are theoretically feasible, this analysis aims to identify the most likely travel mode and route using a few simple assumptions. Namely:

- Foot routes longer than 60 mins are excluded as it is unrealistic
- Where available, car travel is generally preferred over foot and boat.
- The only cases where car is not preferred is:
 - Foot travel is less than 0.5km (500m), meaning that schools are within walking distance.
 - The boat route is significantly faster by more than 20 minutes.
- If no routes are identified for either travel mode, then the point is classified as having no possible routes.

These assumptions are intentionally simple but produce representative estimates of the most practical routes. While real-world travel behaviour is more complex, these estimates are sufficient for this analysis.

Limitations

While OSRM is generally a very powerful routing calculator, it has several limitations.

First, OSRM calculates routes based on data from the OpenStreetMap (OSM), but OSM data can be incomplete in rural areas. OSM is an open-source mapping platform. OSM infrastructure data is compiled through a combination of computer-detected features, publicly available sources, and volunteer contributions. OSRM uses metadata associated with these features (e.g. road surface, connectivity, waterways, distance, and speed limits) to identify feasible routes between two points and select the shortest path. In rural areas, while major features such as the main roads and waterways are generally mapped, much of the detailed metadata is often missing. As a result, routing relies more heavily on assumptions, and some infrastructure may not be captured at all.

Although OSM data is continuously improving, it is important to recognise that coverage and reliability differ significantly between urban and rural settings.

Second, OSRM supports only single-mode routing and does not allow multi-modal journeys. Routes are calculated assuming that the same travel mode (car, foot, or boat) is used for the entire journey. As a result, routes that combine modes (for example, walking to a jetty, traveling by boat, then continuing by car) are not captured. Public transport combined with walking is also not captured here. While additional routes may therefore be theoretically possible, in highly inaccessible areas (that are the focus of this analysis) such combinations would likely remain highly inaccessible in practice, hence it does not materially impact our conclusions.

Third, unlike commercial mapping platforms (e.g. Google Maps, Apple Maps, Waze), OSRM takes a more theoretical approach since it does not incorporate real-world user data that commercial solvers have. Commercial platforms usually adjust routing choices and speeds based on observed user behaviour and live data, including traffic congestion and other potential bottlenecks. OSRM is not designed for this purpose. While this limitation does not significantly impact our conclusions, it is important to note that real-world travel patterns and conditions are inevitably more complex.

That said, OSRM remains a powerful routing engine, particularly suited to performing large-scale routing calculations efficiently. While alternative tools can address some of these limitations, they typically do so at the cost of significantly longer computation times, making them impractical for an analysis of this scale. Such approaches could, however, be explored in future work if needed.

Final Results

Foot Routes

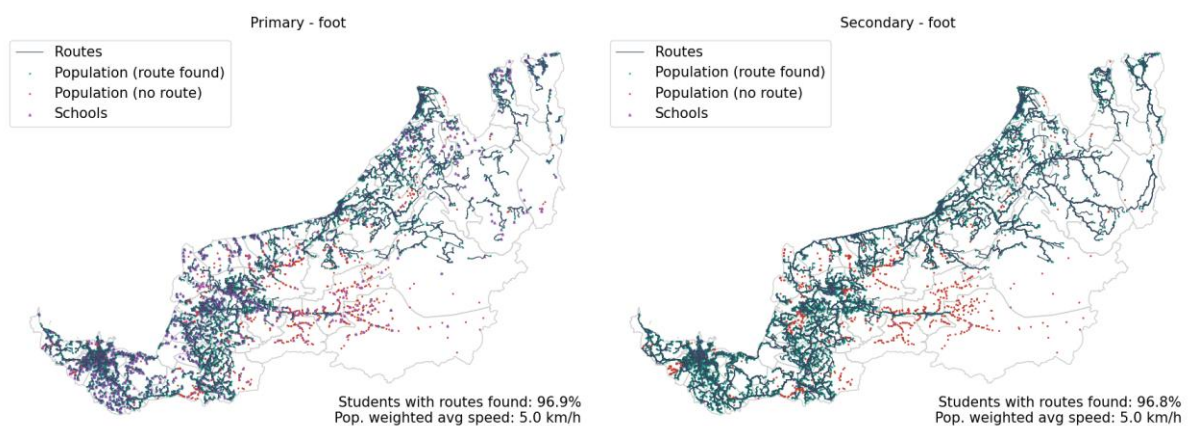


Figure 9: A map showing the final calculated routes for primary and secondary school using the foot profile.

Customised Car Routes

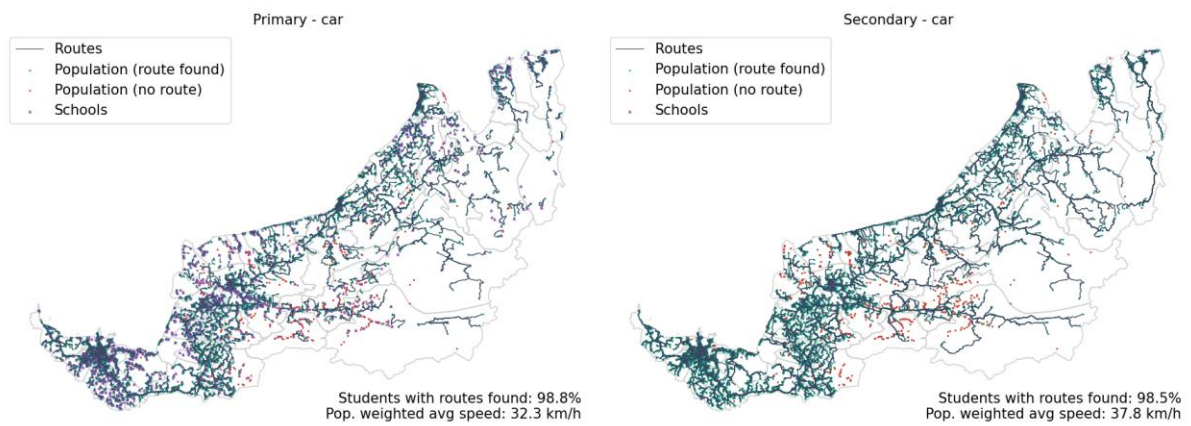


Figure 10: A map showing the final calculated routes for primary and secondary school using the car profile.

Custom Boat Routes

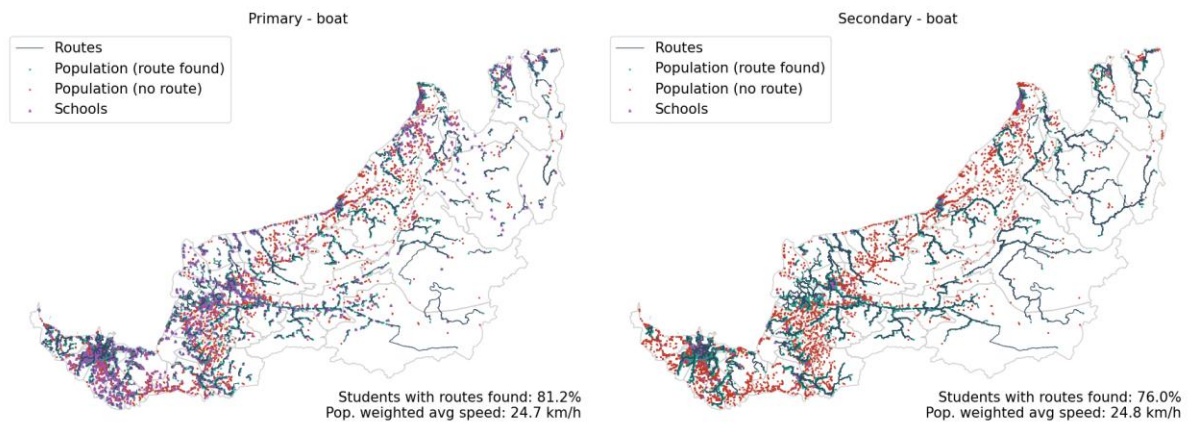


Figure 11: A map showing the final calculated routes for primary and secondary school using the boat profile.

Combined Routes

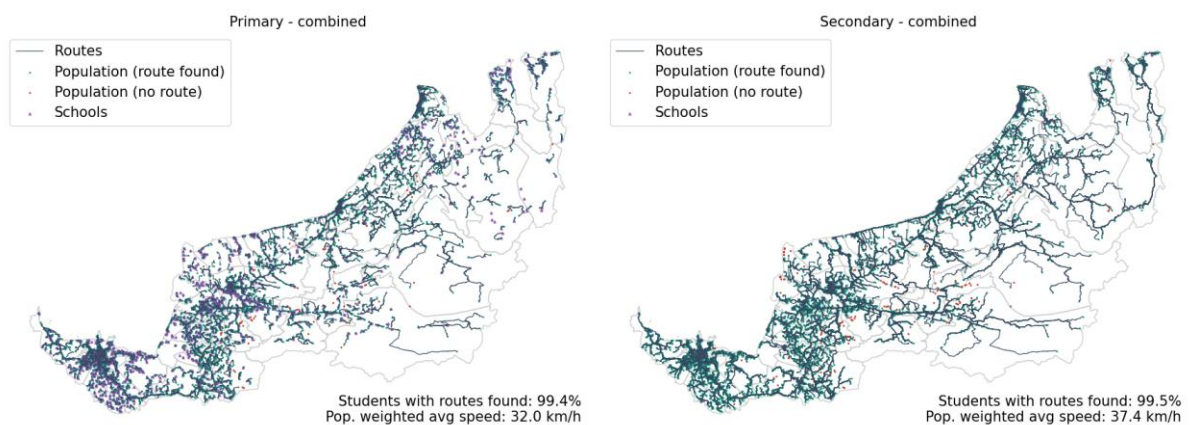


Figure 12: A map the combined routes, showing the most likely routes taken for primary and secondary school.

Population Weighted Average Travel Speeds by District and Travel Mode (km/h)

Table 8: Pop. weighted average travel speeds (km/h) broken down by travel mode and district (primary schools).

District	Foot	Car	Boat	Combined
Overall	5.0	32.3	24.7	32.0
Asajaya	5.0	30.0	22.2	30.0
Bau	5.0	38.4	25.0	38.1
Belaga	5.0	27.9	24.9	27.6
Beluru	5.0	30.7	25.0	30.5
Betong	5.0	32.1	25.0	32.0
Bintulu	5.0	35.4	25.0	35.1
Bukit Mabong	5.0	27.6	25.0	25.5
Dalat	5.0	28.1	25.0	27.9
Daro	5.0	25.4	19.8	24.0
Julau	5.0	34.7	25.0	34.2
Kabong	5.0	33.8	24.9	33.8
Kanowit	5.0	28.8	25.0	28.5
Kapit	5.0	34.3	25.0	42.3
Kuching	5.0	31.7	25.0	31.3
Lawas	5.0	32.2	25.0	32.1
Limbang	5.0	34.3	25.0	34.3
Lubok Antu	5.0	32.7	24.9	32.5
Lundu	5.0	37.9	25.0	37.6
Marudi	5.0	31.4	25.0	31.0
Matu	5.0	27.3	24.8	27.1
Meradong	5.0	33.1	25.0	33.0
Miri	5.0	32.9	25.0	32.3
Mukah	5.0	30.9	25.0	30.8
Pakan	5.0	28.9	25.0	28.9
Pusa	5.0	25.6	25.0	24.8
Samarahan	5.0	39.3	25.0	39.3
Saratok	5.0	28.9	25.6	28.8
Sarikei	5.0	31.0	24.4	31.4
Sebauh	5.0	26.9	25.0	26.8
Selangau	5.0	35.9	25.0	34.6
Serian	5.0	37.8	25.0	36.4
Sibu	5.0	29.8	23.7	28.7
Simunjan	5.0	30.7	24.9	30.3
Song	5.0	28.7	25.0	28.0
Sri Aman	5.0	31.0	25.6	30.7
Subis	5.0	30.6	25.0	30.5
Tanjung Manis	5.0	23.2	21.2	23.2
Tatau	5.0	30.8	25.0	30.4
Tebedu	5.0	30.1	25.0	30.1
Telang Usan	4.9	25.1	25.0	24.9

Table 9: Pop. weighted average travel speeds (km/h) broken down by travel mode and district (secondary schools).

District	Foot	Car	Boat	Combined
Overall	5.0	37.8	24.8	37.4
Asajaya	5.0	34.0	20.4	34.0
Bau	5.0	50.3	25.0	50.3
Belaga	5.0	29.9	25.0	29.5
Beluru	5.0	37.9	25.0	37.9
Betong	5.0	32.3	25.0	32.1
Bintulu	5.0	41.0	25.0	40.9
Bukit Mabong	5.0	30.7	25.0	28.2
Dalat	5.0	35.9	25.0	35.2
Daro	5.0	34.4	24.9	34.2
Julau	5.0	31.8	25.0	31.5
Kabong	5.0	42.6	25.0	42.5
Kanowit	5.0	35.8	25.0	35.0
Kapit	5.0	32.6	25.0	31.5
Kuching	5.0	35.7	25.0	35.4
Lawas	5.0	43.3	25.0	43.4
Limbang	5.0	42.7	25.0	42.4
Lubok Antu	5.0	45.2	25.0	44.2
Lundu	5.0	52.1	25.0	52.1
Marudi	5.0	34.6	25.0	33.7
Matu	5.0	37.1	24.9	35.4
Meradong	5.0	36.5	25.0	35.4
Miri	5.0	35.3	25.0	35.2
Mukah	5.0	37.2	25.0	37.1
Pakan	5.0	33.5	25.0	33.4
Pusa	5.0	32.1	25.0	32.1
Samarahan	5.0	39.8	25.0	39.8
Saratok	5.0	37.6	25.7	37.0
Sarikei	5.0	36.1	24.2	35.5
Sebauh	5.0	35.0	25.0	33.9
Selangau	5.0	51.6	24.8	51.1
Serian	5.0	47.5	25.0	45.7
Sibu	5.0	34.6	23.7	33.8
Simunjan	5.0	45.4	25.0	45.3
Song	5.0	28.8	25.0	28.0
Sri Aman	5.0	38.6	25.0	38.5
Subis	5.0	40.3	25.0	40.2
Tanjung Manis	5.0	24.6	25.0	24.7
Tatau	5.0	41.4	25.0	38.5
Tebedu	5.0	37.1	25.0	37.0
Telang Usan	4.8	33.0	25.0	31.1

Mean, median, Q1, Q3 and IQR of combined route travel times (mins) by district

Table 10: Mean, median, Q1, Q3 and IQR of combined route travel times (mins) by district (primary school).

District	Mean	Median	Q1	Q3	IQR
Overall	5.4	3.1	1.9	5.2	3.3
Asajaya	2.1	1.9	1.1	2.8	1.8
Bau	2.7	2.3	1.4	3.3	1.8
Belaga	18.2	7.1	3.1	19.5	16.4
Beluru	13.9	4.4	1.1	22.1	21.0
Betong	4.6	2.6	1.4	5.1	3.7
Bintulu	6.6	4.6	3.5	7.4	3.9
Bukit Mabong	20.3	9.9	3.7	21.6	17.9
Dalat	3.1	1.4	1.0	2.8	1.9
Daro	3.8	2.3	1.5	3.3	1.8
Julau	7.5	5.8	3.1	9.0	5.9
Kabong	3.6	2.1	1.0	4.3	3.3
Kanowit	5.2	3.3	1.7	6.8	5.2
Kapit	6.4	2.5	1.7	4.5	2.8
Kuching	3.4	3.0	2.1	4.3	2.2
Lawas	3.7	2.3	1.5	3.7	2.2
Limbang	5.7	2.2	1.1	4.3	3.2
Lubok Antu	6.2	3.4	1.4	7.7	6.3
Lundu	5.8	3.3	1.9	6.2	4.3
Marudi	8.5	2.4	0.9	5.4	4.4
Matu	4.4	2.1	1.5	3.6	2.1
Meradong	2.9	2.1	1.2	3.3	2.1
Miri	4.4	3.8	2.4	5.5	3.0
Mukah	8.7	3.6	1.7	8.2	6.4
Pakan	8.6	6.4	3.4	12.1	8.7
Pusa	2.9	1.9	1.1	2.7	1.6
Samarahan	5.0	4.7	2.5	6.6	4.1
Saratok	3.5	2.6	1.4	4.5	3.2
Sarikei	3.2	2.7	1.7	3.7	2.0
Sebauh	23.1	15.7	3.4	29.4	26.0
Selangau	10.9	8.0	3.0	13.1	10.1
Serian	3.6	3.1	1.6	4.6	3.0
Sibu	3.5	2.8	1.8	4.3	2.5
Simunjan	3.5	2.4	1.5	4.5	3.0
Song	5.1	1.8	0.6	5.1	4.5
Sri Aman	5.7	2.8	1.7	5.4	3.8
Subis	14.8	8.3	2.5	19.1	16.6
Tanjung Manis	3.8	2.4	1.5	6.2	4.7
Tatau	13.7	4.6	1.1	19.7	18.6
Tebedu	3.1	2.4	1.3	5.1	3.8
Telang Usan	16.5	2.6	1.2	21.5	20.3

Table 11: Mean, median, Q1, Q3 and IQR of combined route travel times (mins) by district (secondary school).

District	Mean	Median	Q1	Q3	IQR
Overall	13.7	5.3	3.2	11.1	7.9
Asajaya	6.0	5.8	3.7	8.4	4.7
Bau	8.8	7.9	4.0	12.9	8.8
Belaga	41.6	12.3	3.5	58.2	54.7
Beluru	26.7	22.1	12.7	35.3	22.6
Betong	13.3	5.5	3.5	17.4	14.0
Bintulu	9.2	5.2	4.0	8.3	4.3
Bukit Mabong	100.0	79.4	49.1	106.0	56.9
Dalat	20.9	8.1	3.9	12.2	8.3
Daro	20.7	5.7	2.1	49.1	47.0
Julau	36.6	20.7	3.1	47.0	44.0
Kabong	16.0	10.7	6.4	24.3	17.9
Kanowit	16.6	10.9	4.1	23.4	19.4
Kapit	25.0	4.3	3.1	9.0	6.0
Kuching	5.2	4.5	2.9	6.3	3.4
Lawas	29.2	7.1	3.4	16.4	13.1
Limbang	16.6	5.0	3.9	17.4	13.4
Lubok Antu	20.5	11.7	3.9	22.2	18.4
Lundu	16.8	13.7	5.8	24.7	18.9
Marudi	25.7	4.3	2.9	38.1	35.2
Matu	29.8	7.6	4.6	15.5	10.9
Meradong	12.3	3.0	2.3	9.4	7.1
Miri	6.0	4.4	3.0	6.7	3.7
Mukah	35.9	23.9	4.5	58.7	54.3
Pakan	29.3	21.3	5.1	37.5	32.5
Pusa	12.9	4.2	3.2	23.4	20.2
Samarahan	6.5	6.1	3.9	7.8	3.9
Saratok	14.7	3.8	1.7	29.1	27.4
Sarikei	8.7	3.8	2.5	7.3	4.8
Sebauh	66.5	65.3	20.3	95.1	74.9
Selangau	39.6	30.3	20.9	42.1	21.2
Serian	10.1	7.0	4.4	12.1	7.6
Sibu	6.2	3.7	2.3	7.5	5.1
Simunjan	14.1	7.0	4.0	22.7	18.7
Song	28.9	4.4	3.1	19.8	16.7
Sri Aman	18.7	6.2	3.0	31.6	28.7
Subis	33.9	27.8	9.9	47.6	37.7
Tanjung Manis	29.1	35.7	12.8	43.4	30.6
Tatau	54.3	30.3	4.4	73.3	68.8
Tebedu	22.7	20.2	9.5	37.3	27.8
Telang Usan	83.2	70.9	26.6	117.9	91.3

Mean, median, Q1, Q3 and IQR of combined route travel times (km) by district

Table 12: Mean, median, Q1, Q3 and IQR of combined route travel times (km) by district (primary school).

District	Mean	Median	Q1	Q3	IQR
Overall	2.9	1.6	0.8	3.1	2.3
Asajaya	1.2	0.8	0.4	1.4	0.9
Bau	1.7	1.5	0.7	2.2	1.4
Belaga	9.3	3.7	1.1	9.6	8.5
Beluru	7.2	2.9	0.5	12.6	12.1
Betong	2.8	1.3	0.6	2.6	1.9
Bintulu	4.1	2.7	1.7	5.0	3.3
Bukit Mabong	8.0	3.5	0.5	7.3	6.8
Dalat	1.4	0.7	0.4	1.6	1.2
Daro	1.2	0.9	0.5	1.1	0.6
Julau	4.1	3.7	1.3	4.8	3.5
Kabong	2.6	1.1	0.4	3.0	2.6
Kanowit	2.5	1.9	0.8	3.3	2.5
Kapit	2.9	1.6	0.9	2.7	1.7
Kuching	1.8	1.5	0.9	2.4	1.5
Lawas	1.9	1.2	0.7	2.2	1.5
Limbang	3.0	1.2	0.5	3.0	2.4
Lubok Antu	3.5	1.6	0.6	5.1	4.4
Lundu	4.0	2.0	0.8	4.5	3.7
Marudi	4.2	1.3	0.4	3.0	2.6
Matu	2.1	0.8	0.5	1.6	1.1
Meradong	1.7	0.9	0.5	2.0	1.4
Miri	2.5	2.1	1.1	3.3	2.3
Mukah	3.8	1.5	0.7	4.6	3.9
Pakan	4.1	3.4	1.6	5.7	4.1
Pusa	1.0	0.7	0.4	1.1	0.7
Samarahan	3.3	2.9	1.7	4.3	2.6
Saratok	1.8	1.2	0.5	2.2	1.7
Sarikei	1.7	1.3	0.7	2.1	1.4
Sebauh	11.0	6.7	1.6	14.7	13.1
Selangau	6.2	4.9	1.0	9.7	8.7
Serian	2.3	1.6	0.8	2.8	2.0
Sibu	1.8	1.3	0.7	2.2	1.4
Simunjan	1.8	1.1	0.4	2.0	1.6
Song	2.4	0.9	0.2	3.0	2.7
Sri Aman	3.3	1.4	0.7	2.9	2.2
Subis	7.4	4.7	1.0	11.8	10.7
Tanjung Manis	1.5	0.9	0.6	2.4	1.8
Tatau	6.8	2.5	0.3	9.8	9.5
Tebedu	1.6	1.2	0.6	2.6	1.9
Telang Usan	5.7	0.6	0.4	7.8	7.4

Table 13: Mean, median, Q1, Q3 and IQR of combined route travel distance (km) by district (secondary school).

District	Mean	Median	Q1	Q3	IQR
Overall	8.6	3.4	1.8	7.8	6.0
Asajaya	3.6	3.3	1.6	4.4	2.8
Bau	7.7	6.2	3.1	10.7	7.7
Belaga	22.2	5.3	1.8	31.5	29.7
Beluru	16.2	14.5	9.0	20.3	11.3
Betong	7.7	2.6	1.5	12.6	11.2
Bintulu	7.3	3.5	2.4	6.0	3.5
Bukit Mabong	44.7	35.7	24.2	48.2	24.0
Dalat	10.8	5.6	1.8	9.0	7.3
Daro	9.9	2.7	1.6	21.0	19.4
Julau	19.9	10.8	1.2	30.0	28.8
Kabong	12.0	8.8	3.6	20.0	16.4
Kanowit	9.7	6.9	2.1	14.8	12.8
Kapit	11.8	2.6	1.3	5.2	3.9
Kuching	3.2	2.7	1.5	4.1	2.6
Lawas	15.9	5.0	2.2	13.5	11.3
Limbang	11.1	3.6	2.8	12.3	9.6
Lubok Antu	14.1	9.9	3.9	17.5	13.7
Lundu	15.7	12.2	3.6	27.9	24.3
Marudi	13.2	2.1	1.4	20.4	19.0
Matu	14.3	5.4	2.8	10.2	7.4
Meradong	6.5	1.6	1.1	6.5	5.5
Miri	3.8	2.6	1.5	4.3	2.9
Mukah	26.5	11.2	1.8	47.7	45.9
Pakan	16.2	12.5	2.6	19.3	16.6
Pusa	7.6	2.6	1.8	14.2	12.4
Samarahan	4.4	4.1	2.3	5.5	3.2
Saratok	10.7	1.9	1.0	20.7	19.7
Sarikei	5.3	1.9	1.2	5.5	4.3
Sebauh	38.0	41.8	8.3	59.4	51.1
Selangau	29.9	28.8	20.0	34.9	14.9
Serian	7.8	6.2	3.4	10.2	6.9
Sibu	3.7	2.2	1.2	4.2	3.0
Simunjan	11.5	4.6	3.2	19.7	16.5
Song	13.6	1.9	1.2	13.8	12.5
Sri Aman	15.2	3.2	1.4	25.5	24.1
Subis	23.6	19.4	5.5	38.2	32.7
Tanjung Manis	12.0	15.9	5.2	17.8	12.6
Tatau	28.8	18.8	4.2	42.0	37.8
Tebedu	14.5	12.8	6.5	23.9	17.4
Telang Usan	44.9	39.3	11.1	66.7	55.6

Travel mode breakdown of the combined routes by district

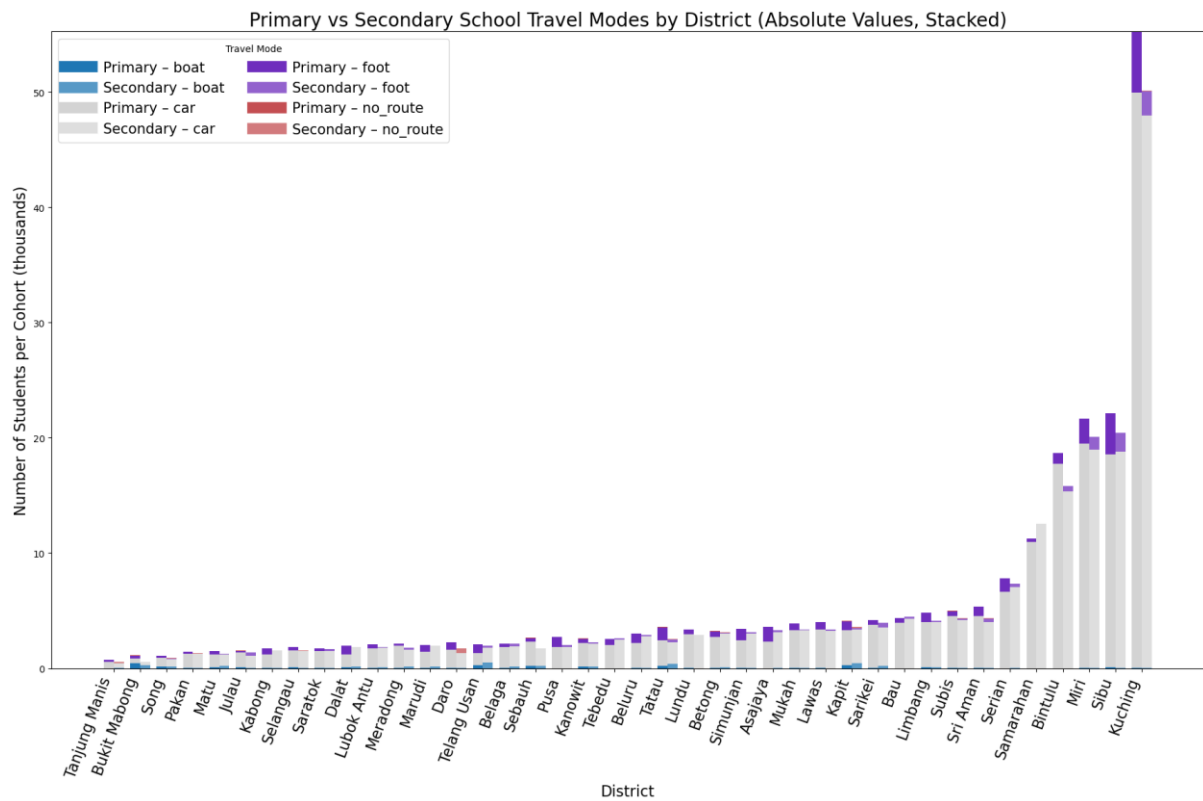


Figure 13: Absolute value breakdown of travel modes in primary and secondary school students, by district.

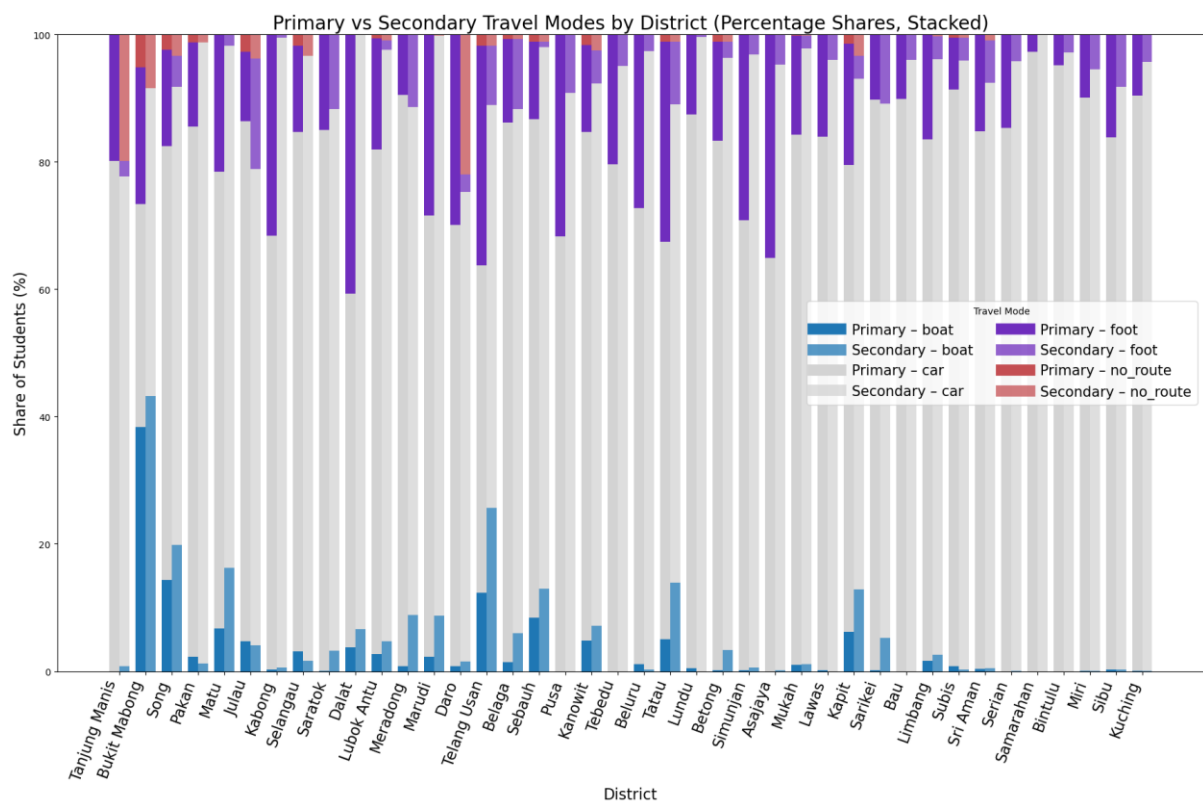


Figure 14: Percentage share breakdown of travel modes in primary and secondary school students, by district.

Travel time breakdown by district

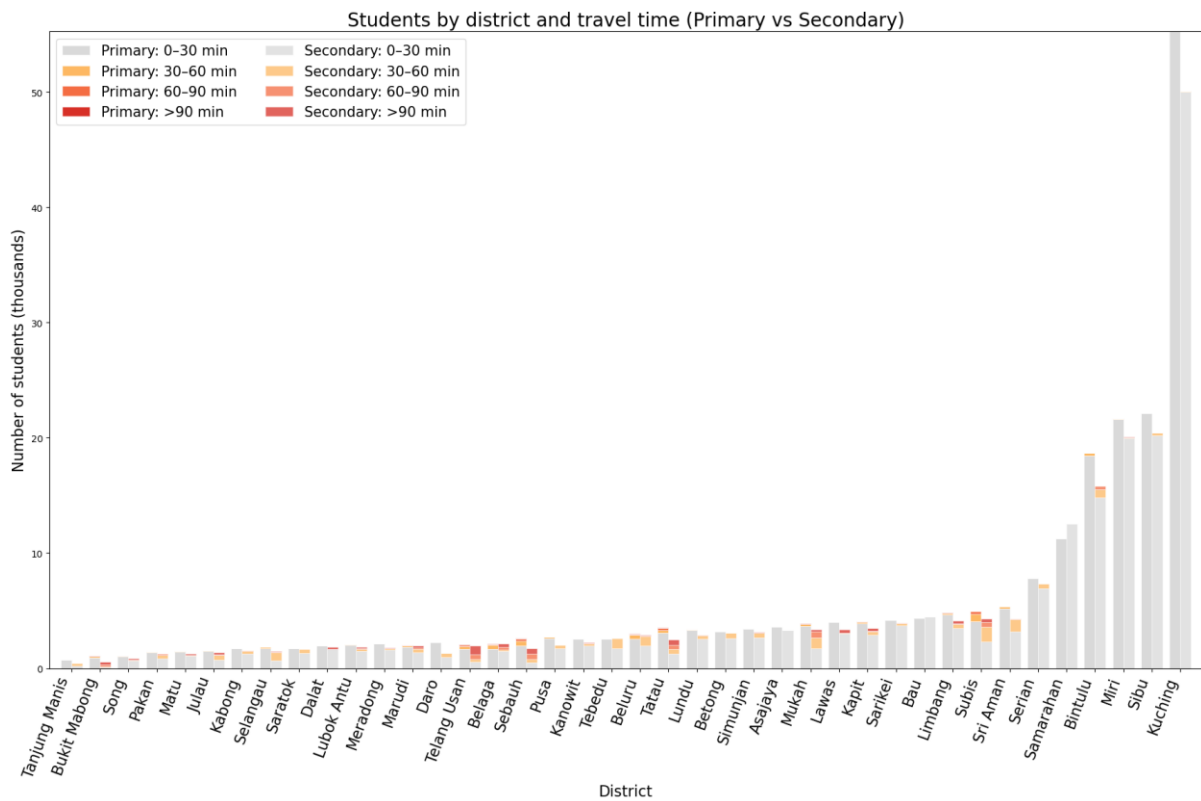


Figure 15: Absolute value breakdown of travel times in primary and secondary school students, by district.

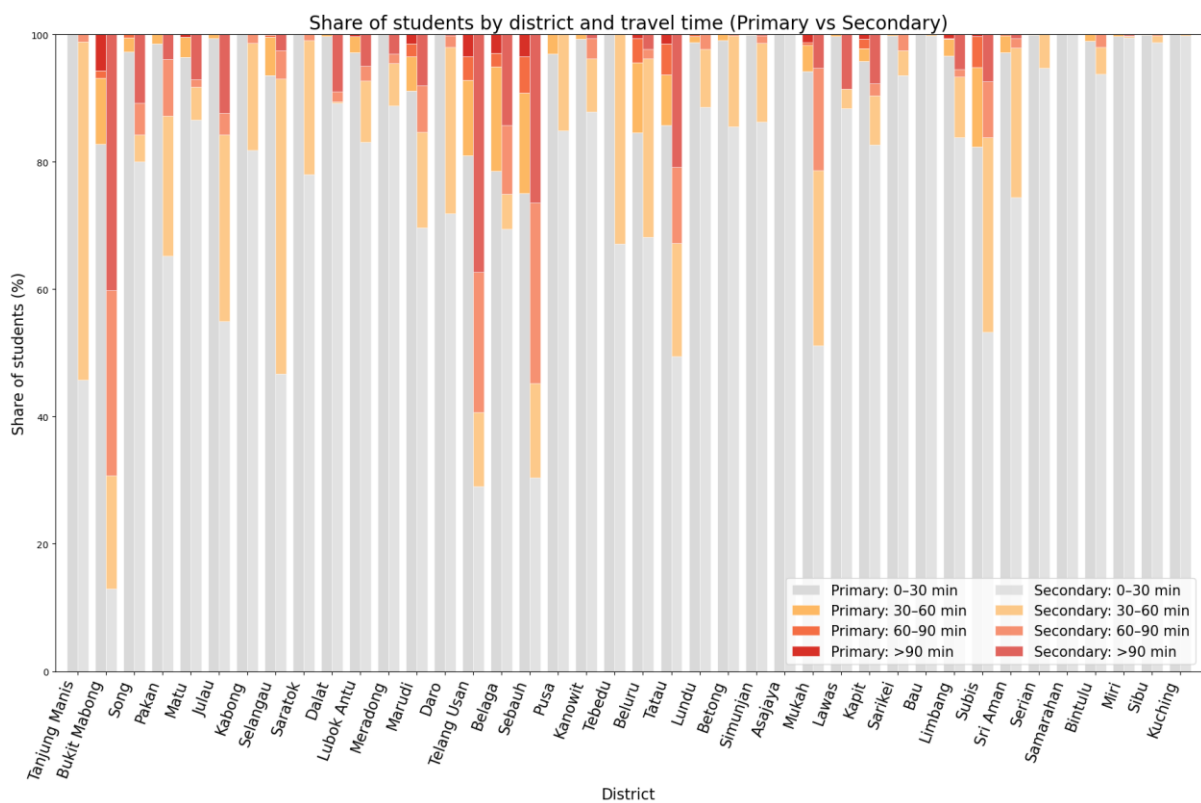


Figure 16: Percentage share breakdown of travel times in primary and secondary school students, by district

Change Log

- 1/2/2026: Technical notes and blogpost first published

Other Links

- [Original blogpost](#)
- [GitHub](#)